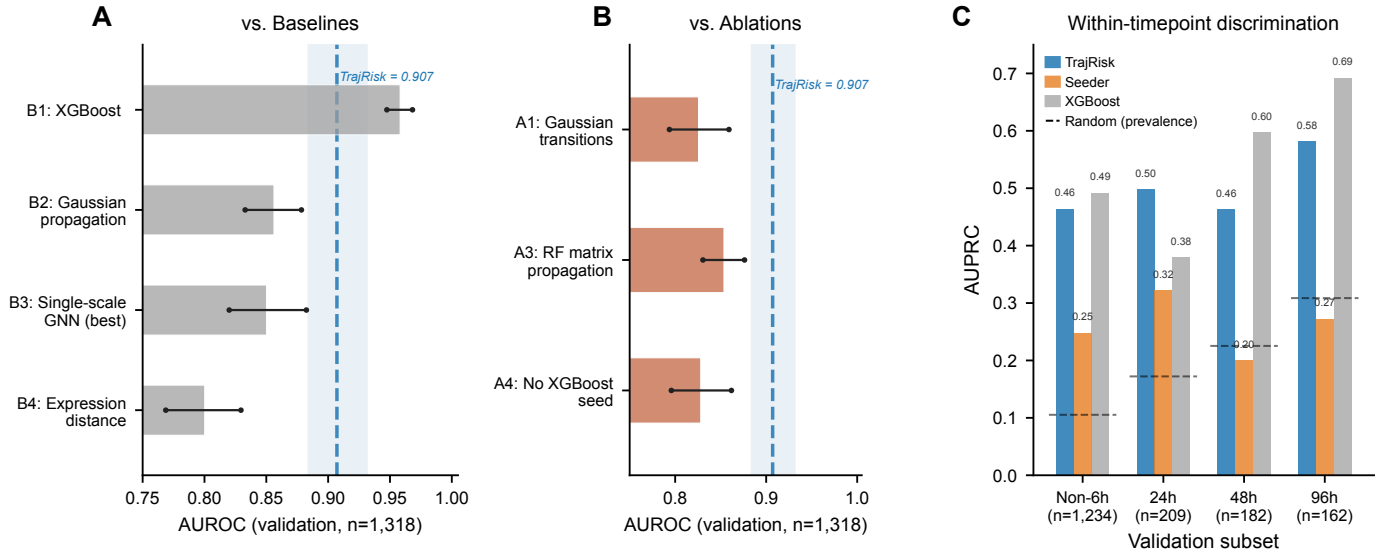

A
vs. Baselines
TrajRisk = 0.907
B1: XGBoost
B2: Gaussian propagation
B3: Single-scale GNN (best)
B4: Expression distance
0.75
0.80
0.85
0.90
0.95
1.00
AUROC (validation, n=1,318)
B
vs. Ablations
TrajRisk = 0.907
A1: Gaussian transitions
A3: RF matrix propagation
A4: No XGBoost seed
0.8
0.9
1.0
AUROC (validation, n=1,318)
C
Within-timepoint discrimination
TrajRisk
Seeder
XGBoost
Random (prevalence)
AUPRC
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.46
0.25
0.49
0.50
0.32
0.38
0.46
0.20
0.60
0.58
0.27
0.69
Non-6h (n=1,234)
24h (n=209)
48h (n=182)
96h (n=162)
Validation subset